Mohit Kumar Meena

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Summary

AI researcher with 5+ years in deep learning research and development, including 3+ years of industrial research at Fujitsu Research of India, leading projects from problem formulation to publication, patent and client proof of concept. Current work centres on reinforcement learning for LLM agent systems, executable environments for training enterprise agents, and efficient learning on graphs with hundreds of millions of nodes. Publications at **AAAI 2026** (first author), **TMLR 2026** and **NeurIPS 2026**, and **three patent applications** filed.

Experience

Fujitsu Research of India

Bengaluru | Jul 2023 – Present

AI Researcher II, promoted from AI Researcher I within 11 months — Jun 2024 – Present

LLM Agents and Multi-Agent Systems

- **Learning how teams of LLM agents should collaborate** — First author • ICLR 2027 submission
  Leading a reinforcement learning method that decides, for each task, how a team of LLM agents divides work and communicates, using execution feedback to recover from failures. Built the multi-GPU training and inference stack on open-weight LLMs, evaluated on competitive programming and math reasoning.
- **Executable replicas of enterprise software for agent training** — Co-author • NeurIPS 2026
  Co-led development of a system that reconstructs closed-source enterprise applications from their manuals and API documentation into runnable environments with databases, APIs and agent tools. Training in these replicas with structured task guidance lifted an 8B open model's task accuracy by **52%**. Built the web application that runs the full generation pipeline.

Efficient and Scalable Graph Learning

- **Adaptive selection and fusion of graph learning models** — First author • AAAI 2026
  Introduced a framework that learns, for every node, which graph neural network architectures to use and how to combine them, pruning weak models during training. Matched or outperformed strong baselines on **16 benchmarks**, with gains of up to **22%** on link prediction. Code released.
- **Memory-efficient training on web-scale graphs** — First author • AAAI 2027 submission
  Leading a theoretically grounded framework that trains graph neural networks on graphs with hundreds of millions of nodes and billions of edges within a fixed GPU memory budget.
- **Matrix sketching for fast graph neural network training** — Co-author • TMLR 2026
  Co-developed a method that compresses the graph before training, with provable optimality for linear models. On a 111M-node graph it trains in under **8 GB** of GPU memory with roughly **339**× faster epochs.

Client Engagements

- **Real-time fraud detection for an enterprise client in Vietnam** — Proof of concept
  Worked on site with the client in Hanoi to define fraud scenarios, data and success criteria, and authored the proposal for a streaming graph AI system (Kafka, Flink, JanusGraph) that flags fraud in real time.
- **Tax-evasion detection on a billion-node graph** — Client deployment
  Delivered an end-to-end pipeline on a heterogeneous graph with billions of nodes, independent of the production graph database. Designed a graph transformer robust to scarce labels and hub nodes, and used edge features for a **3%** downstream gain. Built an interactive viewer for million-node graphs.

Publications and Patents

[C1] **M. Meena**, Y. Punjabi, A. A, V. Sharma, M. Chandran. **Self-Adaptive Graph Mixture of Models.** **AAAI 2026**. Presented at AAAI-26, Singapore. [Paper] [arXiv] [Code]

[J1] A. A, M. Kaul, **M. Meena**, M. Chandran. **Proper Orthogonal Decomposition for Scalable Training of Graph Neural Networks.** **TMLR 2026**. [OpenReview]

[C2] **From Manuals to Worlds: Synthesizing Executable Enterprise Environments for LLM Agents.** **NeurIPS 2026**. Co-author.

[W1] S. A. Nasser, N. C. Kurian, **M. Meena**, S. Shamsi, A. Sethi. **WSSAMNet: Weakly Supervised Semantic Attentive Medical Image Registration Network.** **MICCAI BrainLes Workshop 2022**.

[P1] S. A. Nasser, S. Pathak, K. Singhal, **M. Meena**, N. Gupte, A. Chinmaya, P. Garg, A. Sethi. **Utilizing Radiomic Feature Analysis for Automated MRI Keypoint Detection.** arXiv, 2023. [arXiv]

[P2] S. A. Nasser, **M. Meena**, G. Sresth, A. Sethi. **Leveraging Segmentation to Improve Medical Image Registration.** Preprint, 2023.

Patents: three applications filed through Fujitsu as inventor, on adaptive fusion of graph models, memory-efficient large-graph training and multi-agent LLM collaboration.

Graduate Research, IIT Bombay

- **Master's thesis: Multimodal image registration for image-guided surgery** Advisor: Prof. Amit Sethi
  Developed CNN and Vision Transformer models for registration and segmentation across X-ray, MRI, CT, laparoscopic and ultrasound imaging. Preceding R&D work produced a weakly supervised attention model (W1) that is **6%** more robust than VoxelMorph, the standard learning-based baseline.
- **Low-light image enhancement for driver-assistance systems** Course project • 2021
  Synthesised realistic night-time scenes from the BDD100K driving dataset and trained pix2pix GAN, CAN and U-Net models that clearly outperformed classical CLAHE and Retinex methods.
- **MRI super-resolution with generative adversarial networks** Course project • 2022
  Implemented SRGAN from scratch for 4× super-resolution of brain MRI from the ISBI 2022 BraTSReg challenge, reaching PSNR 47.1 dB and SSIM 0.70.
- **Multimodal brain tumour segmentation** Course project • 2021
  Segmented edema and enhancing and non-enhancing tumour from multi-parametric MRI (BraTS 2018) with U-Net models built from scratch, reaching a Dice score of **0.96** with the 2D variant.

Education

Indian Institute of Technology (IIT) Bombay | M.Tech, CPI **9.29**/10 2021 – 2023

National Institute of Technology (NIT) Uttarakhand | B.Tech, CPI **8.90**/10 2017 – 2021

Technical Skills

LLMs and agents	RL for LLMs (GRPO, REINFORCE), multi-agent systems, MCP, vLLM, Hugging Face
Deep learning	PyTorch, PyTorch Geometric, TensorFlow, graph neural networks, CNNs, ViTs, GANs
Engineering	Python, C++, SQL, multi-GPU training (A100, H100), Docker, FastAPI, Kafka, Flink, Neo4j

Leadership

- **Team lead, Fujitsu Research:** led a team of 3 researchers and 2 interns building the enterprise software replicas used to train and evaluate LLM agents.
- **Mentoring and stakeholder communication:** guided a junior engineer in building an interactive demo of the multi-agent LLM research for Fujitsu leadership.
- **Institute Secretary, Academic Affairs, IIT Bombay:** led a 75+ member council representing 3,300+ Master's students and ran 30+ institute-wide events.